Screenshot of Project

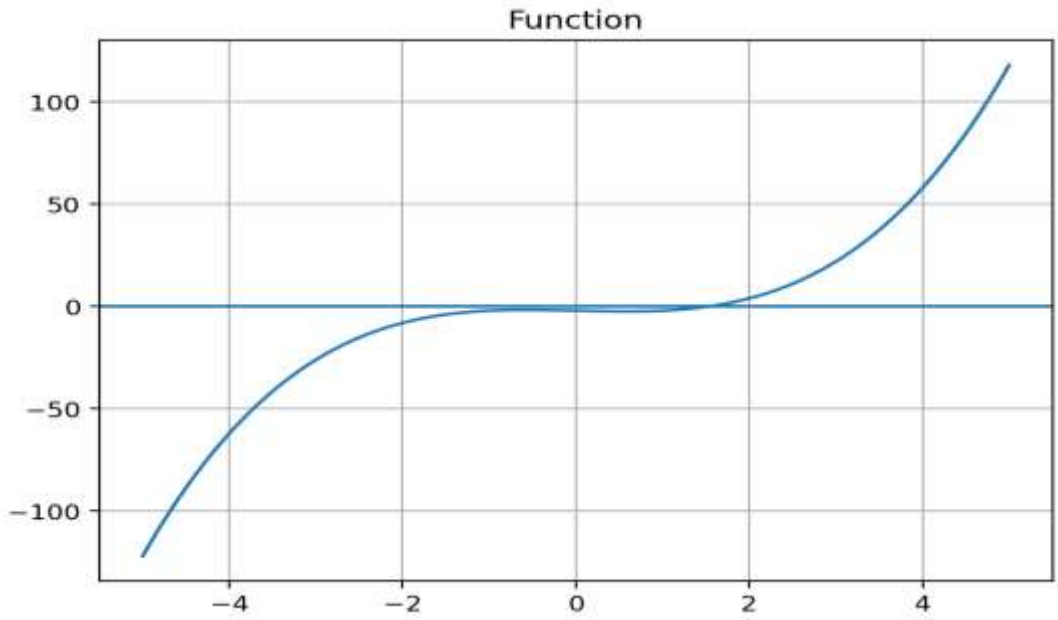
To solve the equation x^3 - x - 2 = 0, we can use numerical methods such as the Newton-Raphson method or the bisection method.

Answer:

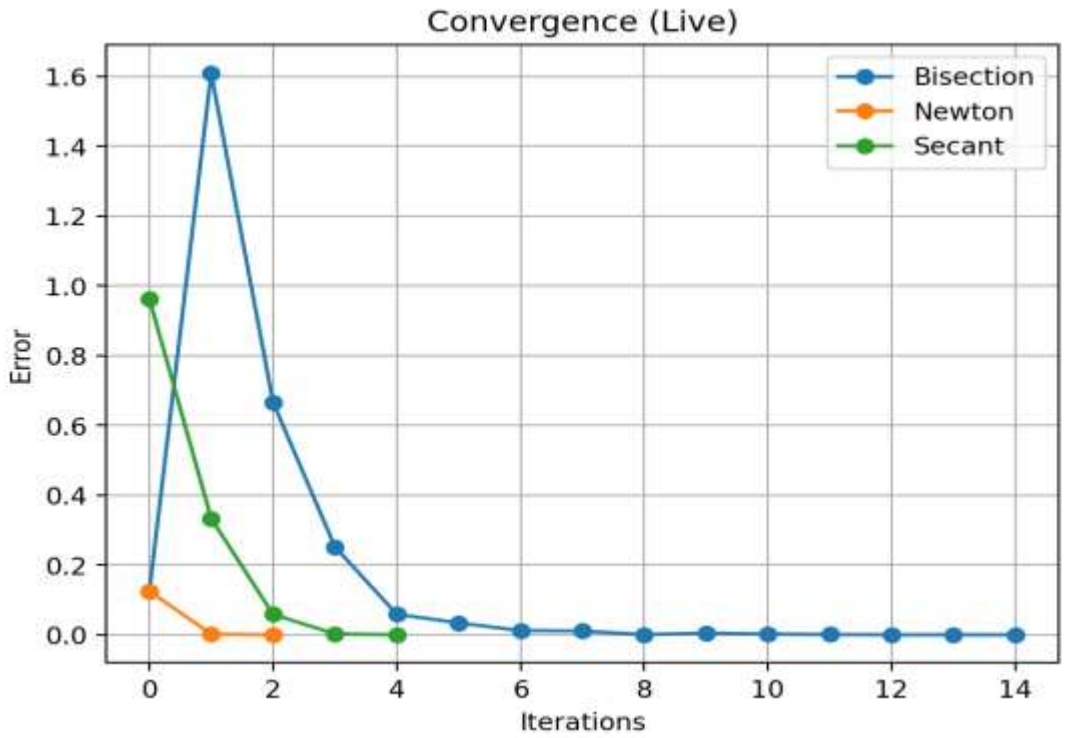
📊 **Algorithm Convergence Visualizer**

🚀 **Real-Time Comparison of Numerical Methods**

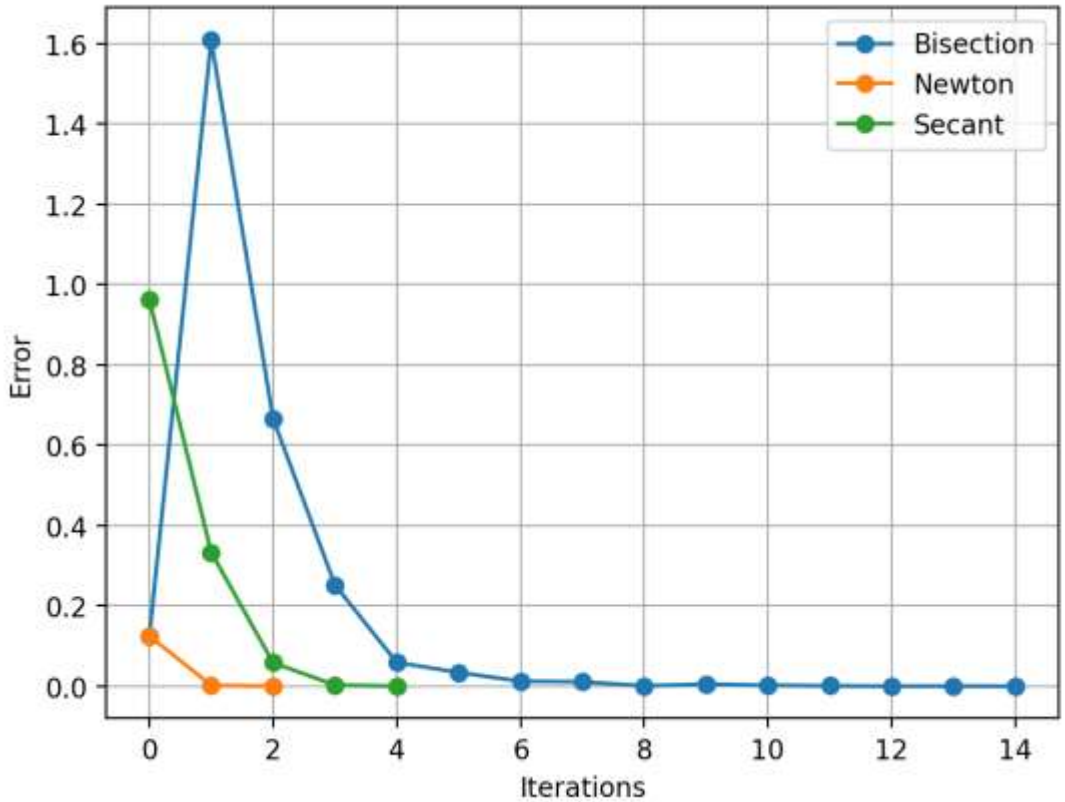
📘 **Function Graph**



📈 **Live Convergence**



📊 **Final Comparison Graph**



Convergence Table

	Bisection	Newton	Secant
0	0.125	0.125	0.963
1	1.6094	0.0021	0.3333
2	0.666	-0.0000005	0.0586
3	0.2522	None	0.0027
4	0.0891	None	0.00002
5	0.0341	None	None
6	0.0123	None	None
7	0.011	None	None
8	0.0006	None	None
9	0.0052	None	None

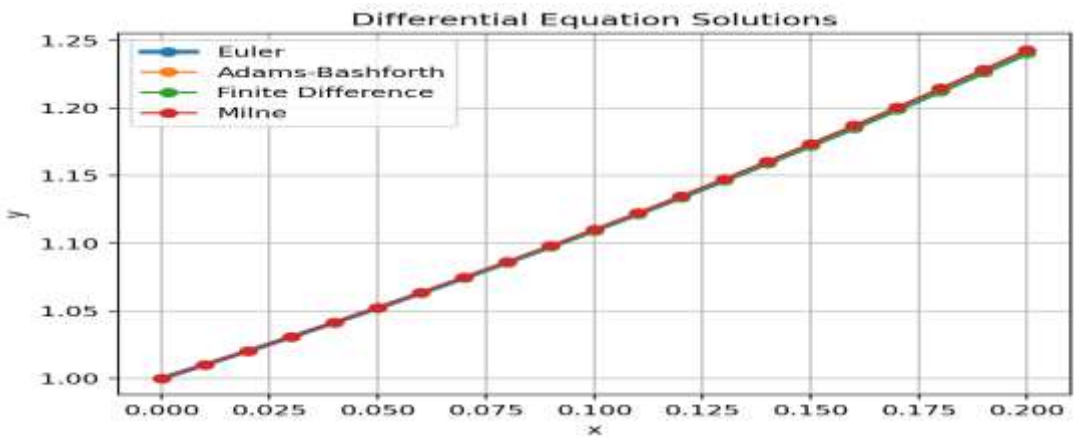
Download Table

Download Graph

Differential Equation

Use this in your app:

dy/dx = x + y
x0 = 0
y0 = 1
h = 0.01
n = 20



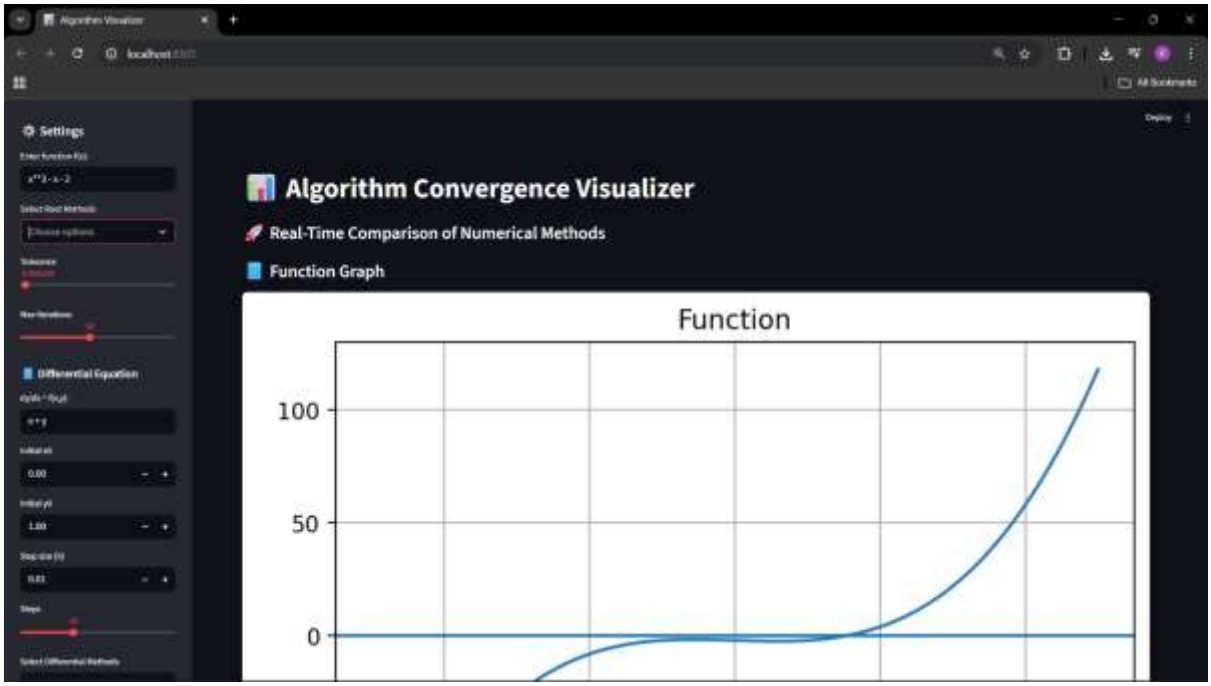
Differential Table

	x	Euler	Adams-Bashforth	Finite Difference	Milne
0	0	1	1	1	1
1	0.01	1.01	1.01	1.01	1.01
2	0.02	1.0202	1.0202	1.0202	1.0202
3	0.03	1.0306	1.0306	1.0306	1.0306
4	0.04	1.0412	1.0413	1.0412	1.0414
5	0.05	1.052	1.0522	1.052	1.0523
6	0.06	1.063	1.0634	1.063	1.0635
7	0.07	1.0743	1.0747	1.0743	1.0748
8	0.08	1.0857	1.0865	1.0857	1.0865
9	0.09	1.0974	1.099	1.0974	1.0991

Download Table

Download Graph

Result:


Settings
Algorithm Convergence Visualizer
Real-Time Comparison of Numerical Methods
Function Graph
Function
100
50
0